# Derivate partiale

• Să se calculeze derivatele parțiale de ordinul I pentru funcțiile:

1. $f(x,y) = e^{x-y^2}$

$\frac{\partial f}{\partial x} = e^{x-y^2}$ ; $\frac{\partial f}{\partial y} = e^{x-y^2} \cdot (-2y)$

2. $f(x,y,z) = e^{x^2+y^2} \cdot \sin^2 z$

$\frac{\partial f}{\partial x} = \sin^2 z \cdot e^{x^2+y^2} \cdot 2x$ ; $\frac{\partial f}{\partial y} = \sin^2 z \cdot e^{x^2+y^2} \cdot 2y$ ;

$\frac{\partial f}{\partial z} = \sin 2z \cdot e^{x^2+y^2}$

3. $f(x,y) = xy \cdot \operatorname{arctg}\left(\frac{x+y}{1-xy}\right)$ ; $xy \neq 1$

$\frac{\partial f}{\partial x} = y \operatorname{arctg}\left(\frac{x+y}{1-xy}\right) + xy \cdot \frac{1}{1+\left(\frac{x+y}{1-xy}\right)^2} \cdot \frac{1-xy+(x+y)y}{(1-xy)^2} =$

$= y \operatorname{arctg}\left(\frac{x+y}{1-xy}\right) + xy \cdot \frac{1+y^2}{(1-xy)^2+(x+y)^2} =$

$= y \operatorname{arctg}\left(\frac{x+y}{1-xy}\right) + xy \cdot \frac{1+y^2}{1-2xy+x^2y^2+x^2+2xy+y^2} =$

$= y \operatorname{arctg}\left(\frac{x+y}{1-xy}\right) + xy \cdot \frac{1+y^2}{(1+x^2)(1+y^2)} =$

$= y \operatorname{arctg}\left(\frac{x+y}{1-xy}\right) + \frac{xy}{1+x^2}$ .

$\frac{\partial f}{\partial y} = x \operatorname{arctg}\left(\frac{x+y}{1-xy}\right) + \frac{xy}{1+y^2}$

4. $f(x,y,z) = x^{y^z}$ ; $x,y > 0$

$\frac{\partial f}{\partial x} = y^z x^{y^z-1}$ ; $\frac{\partial f}{\partial y} = x^{y^z} \cdot \ln x \cdot z \cdot y^{z-1}$ ;

$\frac{\partial f}{\partial z} = x^{y^z} \ln x \cdot y^z \cdot \ln y$ .



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• Să se calculeze derivatele parțiale de ordinul I și II pentru funcțiile:

1. $f(x,y) = e^x \cos y$

I: $\frac{\partial f}{\partial x} = e^x \cos y$ ; $\frac{\partial f}{\partial y} = -e^x \sin y$

II: $\frac{\partial^2 f}{\partial x^2} = e^x \cos y$ ; $\frac{\partial^2 f}{\partial x \partial y} = -e^x \sin y$ ; $\frac{\partial^2 f}{\partial y \partial x} = -e^x \sin y$ ;

$\frac{\partial^2 f}{\partial y^2} = -e^x \cos y$.

2. $f(x,y) = x^3 + xy$

I: $\frac{\partial f}{\partial x} = 3x^2 + y$ ; $\frac{\partial f}{\partial y} = x$

II: $\frac{\partial^2 f}{\partial x^2} = 6x$ ; $\frac{\partial^2 f}{\partial x \partial y} = 1$ ; $\frac{\partial^2 f}{\partial y \partial x} = 1$ ; $\frac{\partial^2 f}{\partial y^2} = 0$

3. $f(x,y) = \frac{x-y}{x+y}$ ; $(x,y) \neq (0,0)$

I: $\frac{\partial f}{\partial x} = \frac{x+y-x+y}{(x+y)^2} = \frac{2y}{(x+y)^2}$ ; $\frac{\partial f}{\partial y} = \frac{-x-y-x+y}{(x+y)^2} = \frac{-2x}{(x+y)^2}$

II: $\frac{\partial^2 f}{\partial x^2} = \frac{0(x+y)^2 - 2y \cdot 2(x+y)}{(x+y)^4} = \frac{-4y}{(x+y)^3}$ ;

$\frac{\partial^2 f}{\partial y \partial x} = \frac{2(x+y)^2 - 2y \cdot 2(x+y)}{(x+y)^4} = \frac{2[x+y-2y]}{(x+y)^3} =$

$= \frac{2(x-y)}{(x+y)^3}$.

$\frac{\partial^2 f}{\partial x \partial y} = \frac{-2(x+y)^2 + 2x \cdot 2(x+y)}{(x+y)^4} = \frac{2(2x-x-y)}{(x+y)^3} =$

$= \frac{2(x-y)}{(x+y)^3}$

$\frac{\partial^2 f}{\partial y^2} = \frac{+2x \cdot 2(x+y)}{(x+y)^4} = \frac{4x}{(x+y)^3}$.



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**4.** $f(x,y) = \arccos \frac{x}{\sqrt{x^2+y^2}}$; $(x,y) \neq (0,0)$

I: $\frac{\partial f}{\partial x} = \frac{-1}{\sqrt{1-\frac{x^2}{x^2+y^2}}} \cdot \frac{\sqrt{x^2+y^2} - x \cdot \frac{2x}{2\sqrt{x^2+y^2}}}{x^2+y^2} =$

$= \frac{-1}{\sqrt{\frac{y^2}{x^2+y^2}}} \cdot \frac{x^2+y^2-x^2}{\sqrt{x^2+y^2}\cdot(x^2+y^2)} = \frac{-y^2}{\sqrt{y^2}(x^2+y^2)} = \frac{-|y|}{x^2+y^2}$

$\frac{\partial f}{\partial y} = \frac{-1}{\sqrt{1-\frac{x^2}{x^2+y^2}}} \cdot \frac{-x \cdot \frac{2y}{2\sqrt{x^2+y^2}}}{x^2+y^2} =$

$= \frac{1}{\sqrt{\frac{y^2}{x^2+y^2}}} \cdot \frac{xy}{\sqrt{x^2+y^2}(x^2+y^2)} = \frac{xy}{\sqrt{y^2}(x^2+y^2)} = \frac{x}{x^2+y^2} \cdot \frac{|y|}{y}$

II: $\frac{\partial^2 f}{\partial x^2} = \frac{0\cdot(x^2+y^2) + |y|\cdot 2x}{(x^2+y^2)^2} = \frac{2x|y|}{(x^2+y^2)^2}$;

$\frac{\partial^2 f}{\partial y \partial x} = \frac{-\frac{1}{2|y|}\cdot 2y(x^2+y^2) + |y|\cdot 2y}{(x^2+y^2)^2} = \frac{2y^3 - x^2y - y^3}{|y|(x^2+y^2)^2} =$

$= \frac{y(y^2-x^2)}{|y|(x^2+y^2)^2}$

$\frac{\partial^2 f}{\partial x \partial y} = \frac{|y|\cdot y(x^2+y^2) - x|y|\cdot 2xy}{y^2(x^2+y^2)^2} = \frac{|y|(x^2+y^2-2x^2)}{y(x^2+y^2)^2} =$

$= \frac{|y|(y^2-x^2)}{y(x^2+y^2)^2}$

$\frac{\partial^2 f}{\partial y^2} = \frac{x\cdot\frac{2y}{2|y|}\cdot y(x^2+y^2) - x|y|(x^2+3y^2)}{y^2(x^2+y^2)^2} =$

$= \frac{xy(x^2+y^2) - xy(x^2+3y^2)}{|y|y^2(x^2+y^2)^2} = \frac{x(x^2+y^2-x^2-3y^2)}{|y|\cdot y(x^2+y^2)^2} =$

$= \frac{-2xy}{|y|(x^2+y^2)^2}$.



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- Să se calculeze derivatele parțiale în punctele specifi

1. $f(x,y) = 2x^2 + xy$ ; $\frac{\partial f}{\partial x}(1,1)$ ; $\frac{\partial f}{\partial y}(3,2)$

$\frac{\partial f}{\partial x} = 4x + y \Rightarrow \frac{\partial f}{\partial x}(1,1) = 4 \cdot 1 + 1 = 5$.

$\frac{\partial f}{\partial y} = x \Rightarrow \frac{\partial f}{\partial y}(3,2) = 3$

2. $f(x,y) = e^{\sin xy}$ ; $\frac{\partial f}{\partial x}\left(1, \frac{\pi}{2}\right)$ ; $\frac{\partial f}{\partial y}(1,0)$

$\frac{\partial f}{\partial x} = e^{\sin xy} \cdot \cos xy \cdot y \Rightarrow \frac{\partial f}{\partial x}\left(1, \frac{\pi}{2}\right) = e \cdot 0 \cdot \frac{\pi}{2} = 0$

$\frac{\partial f}{\partial y} = e^{\sin xy} \cdot \cos xy \cdot x \Rightarrow \frac{\partial f}{\partial y}(1,0) = e^0 \cdot 1 \cdot 1 = 1$

3. $f(x,y) = \sqrt{\sin^2 x + \sin^2 y}$ ; $\frac{\partial f}{\partial x}\left(\frac{\pi}{4}, 0\right)$ ; $\frac{\partial f}{\partial y}\left(\frac{\pi}{4}, \frac{\pi}{4}\right)$

$\frac{\partial f}{\partial x} = \frac{1}{2\sqrt{\sin^2 x + \sin^2 y}} \cdot 2 \sin x \cdot \cos x = \frac{\sin x \cos x}{\sqrt{\sin^2 x + \sin^2 y}} \Rightarrow$

$\Rightarrow \frac{\partial f}{\partial x}\left(\frac{\pi}{4}, 0\right) = \frac{\frac{\sqrt{2}}{2} \cdot \frac{\sqrt{2}}{2}}{\sqrt{\frac{1}{2} + 0}} = \frac{\frac{1}{2}}{\sqrt{\frac{1}{2}}} = \frac{\sqrt{2}}{2}$

$\frac{\partial f}{\partial y} = \frac{1}{2\sqrt{\sin^2 x + \sin^2 y}} \cdot 2 \sin y \cos y = \frac{\sin y \cos y}{\sqrt{\sin^2 x + \sin^2 y}} \Rightarrow$

$\Rightarrow \frac{\partial f}{\partial y}\left(\frac{\pi}{4}, \frac{\pi}{4}\right) = \frac{\frac{\sqrt{2}}{2} \cdot \frac{\sqrt{2}}{2}}{\sqrt{\frac{1}{2} + \frac{1}{2}}} = \frac{1}{2}$

4. $f(x,y) = \ln(1 + x^2 + y^2)$ ; $\frac{\partial f}{\partial x}(1,1)$ ; $\frac{\partial f}{\partial y}(1,1)$

$\frac{\partial f}{\partial x} = \frac{1}{1 + x^2 + y^2} \cdot 2x \Rightarrow \frac{\partial f}{\partial x}(1,1) = \frac{1}{3} \cdot 2 = \frac{2}{3}$

$\frac{\partial f}{\partial y} = \frac{1}{1 + x^2 + y^2} \cdot 2y \Rightarrow \frac{\partial f}{\partial y}(1,1) = \frac{1}{3} \cdot 2 = \frac{2}{3}$



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5. $f(x,y) = \sqrt[3]{x^2y}$ ; $\frac{\partial f}{\partial x}(-2,2)$ ; $\frac{\partial f}{\partial y}(-2,2)$ ; $\frac{\partial^2 f}{\partial x \partial y}(-2,2)$

$\frac{\partial f}{\partial x} = \frac{1}{3\sqrt[3]{(x^2y)^2}} \cdot 2xy \Rightarrow \frac{\partial f}{\partial x}(-2,2) = \frac{1}{3\sqrt[3]{64}} \cdot 2(-4) = \frac{-2}{3}$

$\frac{\partial f}{\partial y} = \frac{1}{3\sqrt[3]{(x^2y)^2}} \cdot x^2 \Rightarrow \frac{\partial f}{\partial y}(-2,2) = \frac{1}{3\sqrt[3]{64}} \cdot 4 = \frac{1}{3}$

$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x}\left[\frac{x^2}{3 \cdot x^{\frac{4}{3}} y^{\frac{2}{3}}}\right] = \frac{\partial}{\partial x}\left(\frac{1}{3} x^{\frac{2}{3}} y^{-\frac{2}{3}}\right) = \frac{1}{3} y^{-\frac{2}{3}} \cdot \frac{2}{3} x^{-\frac{1}{3}} =$

$= \frac{2}{9} x^{-\frac{1}{3}} y^{-\frac{2}{3}} \Rightarrow \frac{\partial^2 f}{\partial x \partial y}(-2,2) = \frac{2}{9} \cdot (-2)^{-\frac{1}{3}} \cdot 2^{-\frac{2}{3}} =$

$= \frac{2}{9} \cdot \frac{1}{\sqrt[3]{-2}} \cdot \frac{1}{\sqrt[3]{4}} = \frac{2}{9} \cdot \frac{1}{-2} = \frac{-1}{9}$

6. $f(x,y) = xy \ln x$ ; $x \neq 0$ ; $\frac{\partial^2 f}{\partial x \partial y}(1,1)$ ; $\frac{\partial^2 f}{\partial y \partial x}(1,1)$

$\frac{\partial f}{\partial x} = y \ln x + xy \cdot \frac{1}{x} = y \ln x + y$

$\frac{\partial f}{\partial y} = x \ln x$

$\frac{\partial^2 f}{\partial x \partial y} = \ln x + x \cdot \frac{1}{x} = \ln x + 1 \Rightarrow \frac{\partial^2 f}{\partial x \partial y}(1,1) = 0 + 1 = 1$

$\frac{\partial^2 f}{\partial y \partial x} = \ln x + 1 \Rightarrow \frac{\partial^2 f}{\partial y \partial x}(1,1) = 0 + 1 = 1$

- Calculați derivatele parțiale de ordinul I pt funcțiile compuse:

1. $f(x,y) = \ln(u^2 + v)$ ; $u(x,y) = e^{x+y^2}$ ; $v(x,y) = x^2 + y$.

$\frac{\partial f}{\partial x} = \frac{\partial f}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial f}{\partial v} \cdot \frac{\partial v}{\partial x}$

$\frac{\partial u}{\partial x} = e^{x+y^2}$ ; $\frac{\partial u}{\partial y} = e^{x+y^2} \cdot 2y$ ; $\frac{\partial v}{\partial x} = 2x$ ; $\frac{\partial v}{\partial y} = 1$



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$$\frac{df}{du} = \frac{1}{u^2+v} \cdot 2u \; ; \; \frac{df}{dv} = \frac{1}{u^2+v}$$

$$\Rightarrow \frac{df}{dx} = \frac{2u}{u^2+v} \cdot e^{x+y^2} + \frac{1}{u^2+v} \cdot 2x = \frac{2}{u^2+v}(u^2+x)$$

$$\frac{df}{dy} = \frac{df}{du} \cdot \frac{du}{dy} + \frac{df}{dv} \cdot \frac{dv}{dy} =$$

$$= \frac{2u}{u^2+v} \cdot 2ye^{x+y^2} + \frac{1}{u^2+v} \cdot 1 = \frac{1}{u^2+v}(4u^2y+1)$$

**2.** $f(x,y) = \operatorname{arctg}\frac{2u}{v}$ ; $u(x,y) = x\sin y$ ; $v(x,y) = x\cos y$.

$$\frac{du}{dx} = \sin y \; ; \; \frac{du}{dy} = x\cos y \; ; \; \frac{dv}{dx} = \cos y \; ; \; \frac{dv}{dy} = -x\sin y$$

$$\frac{df}{du} = \frac{1}{1+\left(\frac{2u}{v}\right)^2} \cdot \frac{2}{v} \; ; \; \frac{df}{dv} = \frac{1}{1+\left(\frac{2u}{v}\right)^2} \cdot \frac{-2u}{v^2} = \frac{-2u}{v^2+4u^2}$$

$$\frac{df}{dx} = \frac{df}{du} \cdot \frac{du}{dx} + \frac{df}{dv} \cdot \frac{dv}{dy} = \frac{2v}{v^2+4u^2}\sin y - \frac{2u}{v^2+4u^2}\cos y$$

$$\frac{df}{dy} = \frac{df}{du} \cdot \frac{du}{dy} + \frac{df}{dv} \cdot \frac{dv}{dy} = \frac{2v}{v^2+4u^2} \cdot x\cos y + \frac{-2u}{v^2+4u^2} - x\sin y =$$

$$= \frac{2v^2}{v^2+4u^2} + \frac{2u^2}{v^2+4u^2} = \frac{2}{v^2+4u^2}(u^2+v^2)$$

**3.** $f(x,y) = f(2x \cdot e^y + 3y\sin 2x)$

Notăm: $u(x,y) = 2xe^y + 3y\sin 2x$

$f(x,y) = f(u)$

$$\frac{\partial f}{\partial x} = f'(u) \cdot \frac{\partial u}{\partial x} = f'(u) \cdot (2e^y + 6y\cos 2x)$$

$$\frac{\partial f}{\partial y} = f'(u) \cdot \frac{\partial u}{\partial y} = f'(u) \cdot (2xe^y + 3\sin 2x)$$



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4. $f(x,y) = \mathcal{F}(u,v,w)$ ; $u(x,y) = xy$ ; $v(x,y) = x^2+y^2$ ; $w(x,y) = 2x+3y$

$$\frac{\partial f}{\partial x} = \frac{\partial \mathcal{F}}{\partial u}\cdot\frac{\partial u}{\partial x} + \frac{\partial \mathcal{F}}{\partial v}\cdot\frac{\partial v}{\partial x} + \frac{\partial \mathcal{F}}{\partial w}\cdot\frac{\partial w}{\partial x} =$$

$$= y\frac{\partial \mathcal{F}}{\partial u} + 2x\frac{\partial \mathcal{F}}{\partial v} + 2\frac{\partial \mathcal{F}}{\partial w}$$

$$\frac{\partial f}{\partial y} = \frac{\partial \mathcal{F}}{\partial u}\cdot\frac{\partial u}{\partial y} + \frac{\partial \mathcal{F}}{\partial v}\cdot\frac{\partial v}{\partial y} + \frac{\partial \mathcal{F}}{\partial w}\cdot\frac{\partial w}{\partial y} =$$

$$= x\cdot\frac{\partial \mathcal{F}}{\partial u} + 2y\frac{\partial \mathcal{F}}{\partial v} + 3\frac{\partial \mathcal{F}}{\partial w}$$

- Calculați derivatele parțiale de ordinul II : $\frac{\partial^2 f}{\partial x^2}$ ; $\frac{\partial^2 f}{\partial x \partial y}$ și $\frac{\partial^2 f}{\partial y^2}$ pentru funcțiile compuse.

1. $f(x,y) = \mathcal{F}(u,v)$ ; $u(x,y) = x+y$ ; $v(x,y) = x^2+y^2$.

$$\frac{\partial f}{\partial x} = \frac{\partial \mathcal{F}}{\partial u}\cdot\frac{\partial u}{\partial x} + \frac{\partial \mathcal{F}}{\partial v}\cdot\frac{\partial v}{\partial x} = \frac{\partial \mathcal{F}}{\partial u} + 2x\frac{\partial \mathcal{F}}{\partial v}$$

$$\frac{\partial f}{\partial y} = \frac{\partial \mathcal{F}}{\partial u}\cdot\frac{\partial u}{\partial y} + \frac{\partial \mathcal{F}}{\partial v}\cdot\frac{\partial v}{\partial y} = \frac{\partial \mathcal{F}}{\partial u} + 2y\frac{\partial \mathcal{F}}{\partial v}$$

Notăm: $\frac{\partial \mathcal{F}}{\partial u} = g(u,v)$ ; $\frac{\partial \mathcal{F}}{\partial v} = h(u,v)$

$$\frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial}{\partial x}\left(g(u,v) + 2x\,h(u,v)\right) =$$

$$= \frac{dg}{\partial u}\cdot\frac{\partial u}{dx} + \frac{\partial g}{\partial v}\cdot\frac{dv}{\partial x} + 2h(u,v) + 2x\left(\frac{\partial h}{\partial u}\cdot\frac{\partial u}{\partial x} + \frac{\partial h}{\partial v}\cdot\frac{\partial v}{\partial x}\right) =$$

$$= \frac{\partial g}{\partial u} + \frac{\partial g}{\partial v}\cdot 2x + 2\frac{\partial \mathcal{F}}{\partial v} + 2x\left(\frac{\partial h}{\partial u}\cdot + \frac{\partial h}{\partial v}\cdot 2x\right) =$$

$$\frac{\partial g}{\partial u} = \frac{\partial}{\partial u}(g) = \frac{\partial}{\partial u}\left(\frac{\partial \mathcal{F}}{\partial u}\right) = \frac{\partial^2 \mathcal{F}}{\partial u^2}$$

$$\frac{\partial g}{\partial v} = \frac{\partial}{\partial v}(g) = \frac{\partial}{\partial v}\left(\frac{\partial \mathcal{F}}{\partial u}\right) = \frac{\partial^2 \mathcal{F}}{\partial v \partial u}$$



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$$\frac{\partial h}{\partial u} = \frac{\partial}{\partial u}(h) = \frac{\partial}{\partial u}\left(\frac{\partial f}{\partial v}\right) = \frac{\partial^2 f}{\partial u \partial v}$$

$$\frac{\partial h}{\partial v} = \frac{\partial}{\partial v}\left(h\right) = \frac{\partial}{\partial v}\left(\frac{\partial f}{\partial v}\right) = \frac{\partial^2 f}{\partial v^2}$$

$$\Rightarrow \frac{\partial^2 f}{\partial x^2} = \frac{\partial^2 g}{\partial u^2} + 2x \cdot \frac{\partial^2 g}{\partial v \partial u} + 2\frac{\partial f}{\partial v} + 2x\frac{\partial^2 g}{\partial u \partial v} + 4x\frac{\partial^2 g}{\partial v^2}.$$

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial}{\partial x}\left(g(u,v) + 2y\,h(u,v)\right) =$$

$$= \frac{\partial g}{\partial u}\cdot\frac{\partial u}{\partial x} + \frac{\partial g}{\partial v}\cdot\frac{\partial v}{\partial x} + 2y\left(\frac{\partial h}{\partial u}\cdot\frac{\partial u}{\partial x} + \frac{\partial h}{\partial v}\cdot\frac{\partial v}{\partial x}\right) =$$

$$= \frac{\partial^2 g}{\partial u^2}\cdot + \frac{\partial^2 g}{\partial v \partial u}\cdot 2x + 2y\cdot\frac{\partial^2 g}{\partial u \partial v}\cdot + 4xy\frac{\partial^2 g}{\partial v^2}$$

$$\frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial}{\partial y}\left(g(u,v) + 2y\,h(u,v)\right) =$$

$$= \frac{\partial g}{\partial u}\cdot\frac{\partial u}{\partial y} + \frac{\partial g}{\partial v}\cdot\frac{\partial v}{\partial y} + 2h(u,v) + 2y\left(\frac{\partial h}{\partial u}\cdot\frac{\partial u}{\partial y} + \right.$$

$$\left. + \frac{\partial h}{\partial v}\cdot\frac{\partial v}{\partial y}\right) =$$

$$= \frac{\partial^2 g}{\partial u^2} + \frac{\partial^2 g}{\partial v \partial u}\cdot 2y + 2\cdot\frac{\partial g}{\partial v} + 2y\frac{\partial^2 g}{\partial u \partial v} + 4y^2\frac{\partial^2 g}{\partial v^2}.$$

• Arătați că funcțiile următoare verifică ecuațiile indicate:

1. $f(x,y) = g\left(\frac{y}{x}\right)$, verifică ecuația $x\frac{\partial f}{\partial x} + y\frac{\partial f}{\partial y} = 0$

Notăm: $u(x,y) = \frac{y}{x} \Rightarrow f(x,y) = g(u)$

$$\frac{\partial f}{\partial x} = g'(u)\cdot\frac{\partial u}{\partial x} = g'(u)\cdot\frac{-y}{x^2}$$

$$\frac{\partial f}{\partial y} = g'(u)\cdot\frac{\partial u}{\partial y} = g'(u)\cdot\frac{1}{x}$$



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$$\Rightarrow x \cdot \frac{\partial f}{\partial x} + y \cdot \frac{\partial f}{\partial y} = x \cdot f'(u) \cdot \frac{-y}{x^2} + y \cdot f'(u) \cdot \frac{1}{x} = 0$$

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**2.** $f(x,y) = y \cdot f(x^2 - y^2)$, verifică ecuația:

$$\frac{1}{x} \cdot \frac{\partial f}{\partial x} + \frac{1}{y} \cdot \frac{\partial f}{\partial x} - \frac{1}{y^2} f(x,y)$$

Notăm: $u(x,y) = x^2 - y^2$

$$\frac{\partial f}{\partial x} = y \cdot f'(u) \cdot \frac{\partial u}{\partial x} = 2xy\, f'(u)$$

$$\frac{\partial f}{\partial y} = f(u) + y f'(u) \frac{\partial u}{\partial y} = f(u) - 2y^2 f'(u)$$

$$\Rightarrow \frac{1}{x} \cdot \frac{\partial f}{\partial x} + \frac{1}{y} \cdot \frac{\partial f}{\partial x} = \frac{1}{x} \cdot 2xy\, f'(u) + \frac{1}{y}\left(f(u) - 2y^2 f'(u)\right) -$$

$$= \cancel{2y f'(u)} + \frac{1}{y} f(u) - \cancel{2y f'(u)} =$$

$$= \frac{1}{y} f(u) = \frac{1}{y^2} f(x,y) = \frac{1}{y^2} \cdot y f(u) = \frac{1}{y} f(u)$$

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